

GETTING TIME FROM NTP SERVER

Made Easy

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Every once in a while, you come across an idea where keeping time is a prime concern. For example, imagine a relay that has to be activated at a certain time, or a data logger that has to store values at precise intervals.

The first thing that comes to our mind is to use a real-time clock (RTC) chip. But since these chips are not accurate, we need to keep doing manual adjustments to keep them synchronised.

The solution here is to use network time protocol (NTP). The

ESP8266 NodeMCU has access to the internet, so you can get day of the week and precise time within a few milliseconds of Coordinated Universal Time (UTC) for free! And you don't need any additional hardware circuitry.

NTP is a standard internet protocol (IP) for synchronising computer clocks to some reference over a network. The protocol can be used to synchronise all networked devices to UTC within a few milliseconds—50 milliseconds over the public internet and under 5 milliseconds in a LAN

environment. UTC is a world-wide time standard, closely related to Greenwich Mean Time (GMT). UTC is the same worldwide.

NTP sets the clocks of computers to UTC and any local time zone offset, or daylight saving time offset, is applied by the client. In this manner clients can synchronise to servers regardless of location and time zone differences.

NTP can operate in a number of ways. The most common configuration is to operate in client-server mode. The client device, such as an ESP8266, connects to the server using the User Datagram Protocol (UDP) on port 123. A client then transmits a request packet to an NTP server.

In response to this request the NTP server sends a time stamp packet. A time stamp packet contains multiple information like UNIX time stamp, accuracy, delay, or time zone. A client can then parse out current day of the week and time values.

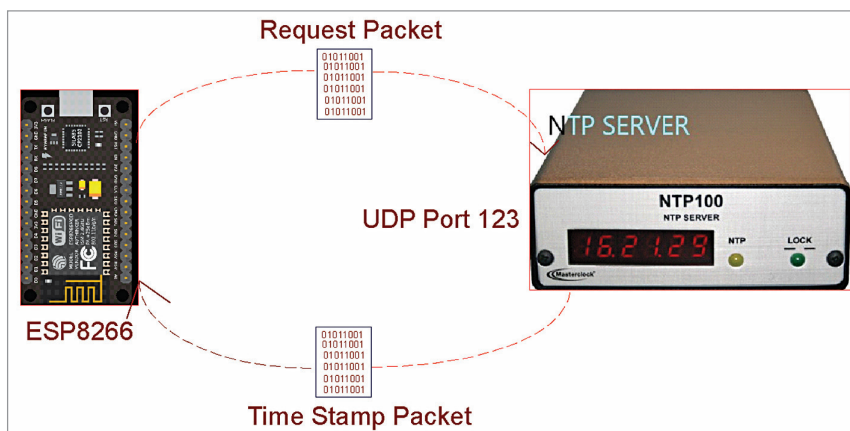


Fig. 1: NTP communication format structure

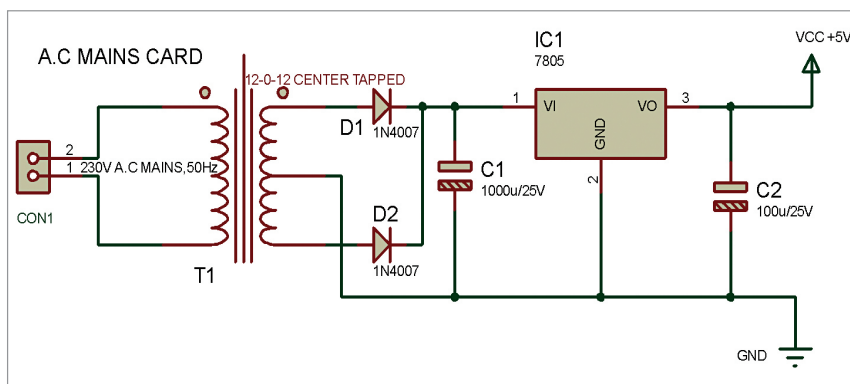


Fig. 2: Regulated power supply

The circuit and working

The structure of NTP protocol is shown in Fig. 1. The NTP communication format structure comprises a regulated power supply and the main circuit.

The regulated power supply, shown in Fig. 2, is built around a 12V-0-12V, 500mA step-down centre-tapped transformer (X1), full-wave rectifier with 1N4007 diodes (D1 and D2), 5V regulator 7805 (IC1), 1000μF, 25V electrolytic capacitors (C1) and 100μF, 25V electrolytic capacitor (C2). Alternatively,